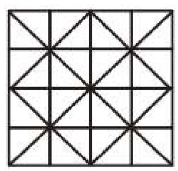
COUNTING FIGURES

(PRACTICE SET)

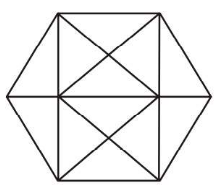
PIYUSH VARSHNEY SIR

1. Find out the number of squares in the given pattern.



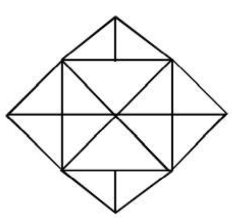
(a) 26 (b) 30
(c) 35 (d) 38

2. How many triangles are there in the following figure ?



(a) 20 (b) 24
(c) 28 (d) 32

3. How many triangles are there in the given figure ?

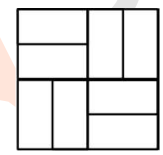


(a) 18 (b) 28
(c) 20 (d) 24

4. How many triangles are there in the given figure ?

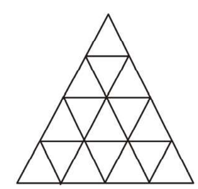
(a) 7 (b) 5
(c) 6 (d) 9

5. How many rectangles are there in the given figure ?



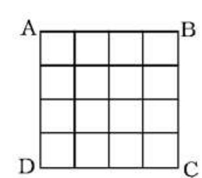
(a) 24 (b) 16
(c) 21 (d) 14

6. How many triangles are there in the following figure?



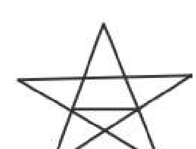
(a) 29 (b) 27
(c) 23 (d) 30

7. How many squares are there in the square figure ABCD ?



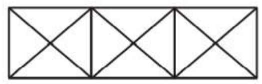
(a) 16 (b) 17
(c) 26 (d) 30

8. How many triangles are there in the given diagram?



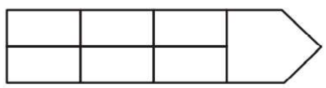
(a) 6 (b) 10
(c) 12 (d) 14

9. How many triangles are there in the given figure ?



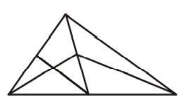
(a) 18 (b) 24
(c) 28 (d) 30

10. How many rectangles are there in the given diagram?



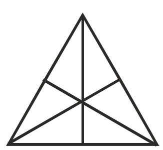
(a) 4 (b) 7
(c) 9 (d) 18

11. How many triangles are there in the following figure ?



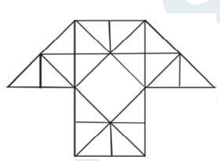
(a) 11 (b) 13
(c) 9 (d) 15

12. How many triangles are there in the following figure ?



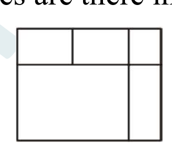
(a) 16 (b) 13
(c) 9 (d) 7

13. How many triangles are there in the given figure ?



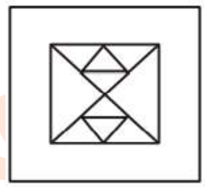
(a) 29 (b) 38
(c) 40 (d) 35

14. How many rectangles are there in the figure given ?



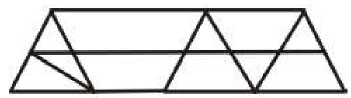
(a) 8 (b) 9
(c) 10 (d) 11

15. How many triangles are there in this figure ?



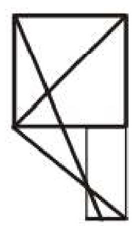
(a) 12 (b) 14
(c) 16 or more (d) 10

16. Count the number of triangles in the figure below and select the correct answer from the response.



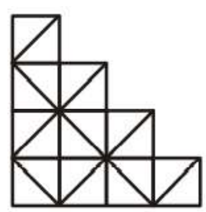
(a) 7 (b) 8
(c) 9 (d) 11

17. How many triangles in all can be found in the following figure?



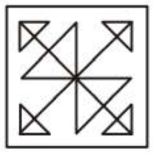
(a) 12 (b) 11
(c) 15 (d) 13

18. How many squares are there in the given figure?



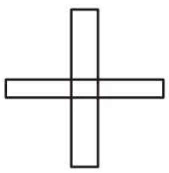
(a) 10 (b) 11
(c) 12 (d) 14

19. Find out the number of triangles in this figure.



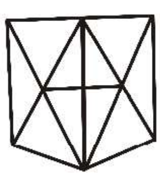
(a) 12 (b) 14
(c) 16 (d) 18

20. How many rectangles are formed in the figure given below ?



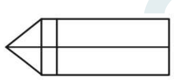
(a) 10 (b) 11
(c) 12 (d) 13

21. How many triangles are there in the following figure?



(a) 23 (b) 24
(c) 18 (d) 20

22. How many rectangles are there in the following figure?

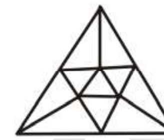


(a) 7 (b) 6
(c) 8 (d) 9

23. How many triangles are there in the given figure?

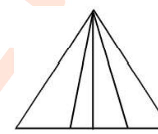
(a) 10 (b) 12
(c) 14 (d) 16

24. How many triangles are there in the given figure?



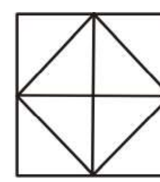
(a) 16 (b) 15
(c) 14 (d) 13

25. How many triangles are there in the given figure?



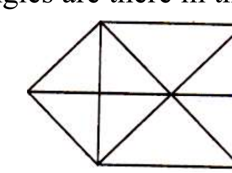
(a) 5 (b) 12
(c) 9 (d) 10

26. How many triangles are there in the following figure ?



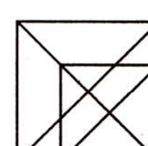
(a) 8 (b) 10
(c) 12 (d) 14

27. How many triangles are there in the given figure?



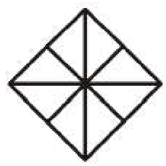
(a) 15 (b) 16
(c) 17 (d) 18

28. How many triangles are there in the given figure?



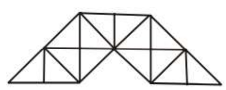
(a) 16 (b) 18
(c) 21 (d) 20

29. How many rhombuses are there in the given diagram?



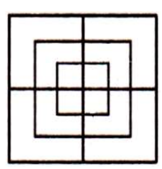
(a) 4 (b) 1
(c) 5 (d) 6

30. Count the number of triangles in the following figure.



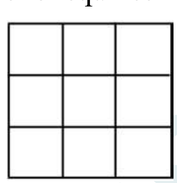
(a) 27 (b) 23
(c) 29 (d) 31

31. How many squares are there in this figure ?



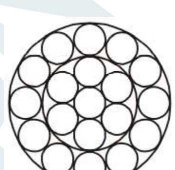
(a) 8 (b) 12
(c) 15 (d) 18

32. The maximum number of squares in the given figure is?



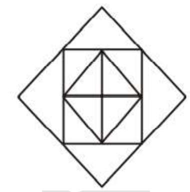
(a) 9 (b) 10
(c) 13 (d) 14

33. How many circles are there in this figure ?



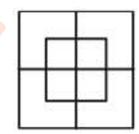
(a) 19 (b) 18
(c) 17 (d) 21

34. How many squares are there in this figure ?



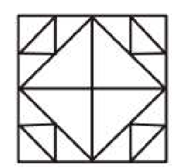
(a) 4 (b) 7
(c) 6 (d) 8

35. How many squares are there in the given figure?



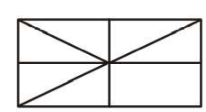
(a) 7 (b) 12
(c) 8 (d) 10

36. How many triangles are there in this figure ?



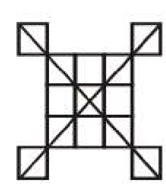
(a) 24 (b) 26
(c) 28 (d) 20

37. How many triangles are there in the figure below ?



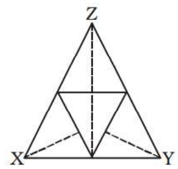
(a) 8 (b) 10
(c) 12 (d) 11

38. Find out the number of squares in the given figure.



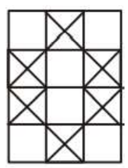
(a) 13 (b) 14
(c) 17 (d) 18

39. Find the number of triangles in the given figure.



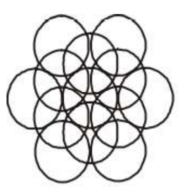
(a) 17 (b) 15
(c) 13 (d) 9

40. Find out the number of squares in the given pattern.



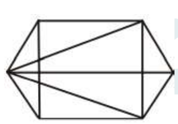
(a) 20 (b) 23
(c) 12 (d) 18

41. How many circles are there in the following figure ?



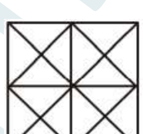
(a) 12 (b) 13
(c) 14 (d) 15

42. Find the number of triangles in the given figure :



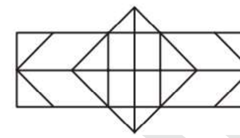
(a) 11 (b) 14
(c) 16 (d) 22 or more

43. How many triangles are there in the given figure ?



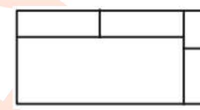
(a) 40 or more (b) 16
(c) 18 (d) 28

44. How many rectangles are there in the given figure?



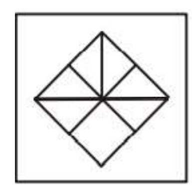
(a) 8 (b) 15
(c) 24 (d) 30

45. How many rectangles are there in the question figure ?



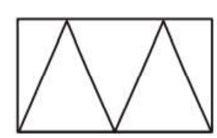
(a) 6 (b) 7
(c) 8 (d) 9

46. How many triangles are there in the given figure ?



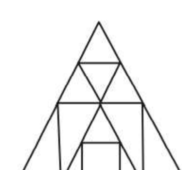
(a) 7 (b) 10
(c) 8 (d) 9

47. How many triangles are there in the given figure?



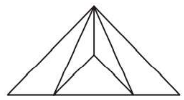
(a) 5 (b) 7
(c) 8 (d) 9

48. Find out the number of triangles in the given figure.



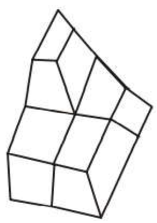
(a) 13 (b) 15
(c) 16 (d) 17

49. Find the number of triangles in the given figure:



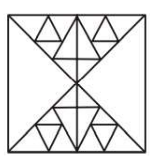
(a) 6 (b) 7
(c) 8 (d) 9

50. The figure below is a drawing of a pile of blocks. When taken apart, how many blocks would be there ?



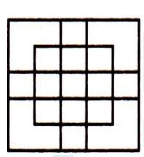
(a) 6 (b) 3
(c) 4 (d) 5

51. Find out the number of triangles in the given figure.



(a) 34 (b) 38
(c) 44 (d) 48 or more

52. How many squares are there in this figure ?

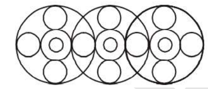


(a) 18 (b) 19
(c) 25 (d) 27

53. How many triangles are there in the given figure ?

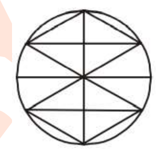
(a) 11 (b) 12 or more
(c) 9 (d) 10

54. Find out the number of circles in the given figure.



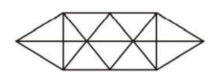
(a) 18 (b) 19
(c) 16 (d) 20

55. How many triangles are embedded in the figure given below ?



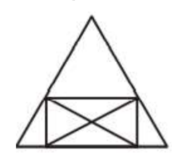
(a) 16 (b) 6
(c) 22 (d) 24

56. How many triangles are there in the figure?



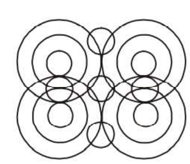
(a) 24 (b) 14
(c) 28 (d) 20

57. Find the number of triangles in the following figure :



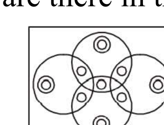
(a) 14 (b) 10
(c) 12 (d) 8

58. Find out the number of circles in the given figure :



(a) 14 (b) 16
(c) 17 (d) 18

59. How many circles are there in this figure ?



(a) 16 (b) 13
(c) 17 (d) 22

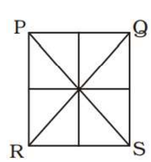
60. How many faces can you count in this 3 dimensional model ?

(a) 12 (b) 14
(c) 16 (d) 18

61. The number of triangles in the following diagram is :

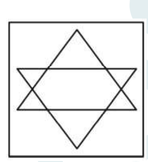
(a) 13 (b) 14
(c) None (d) 17

62. How many quadrilaterals are there in the following figure?



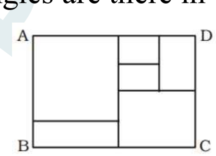
(a) 6 (b) 7
(c) 8 (d) 9

63. How many triangles are there in the following square ?



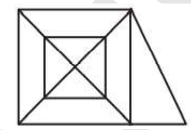
(a) 8 (b) 7
(c) 9 (d) 6

64. How many rectangles are there in the figure ABCD ?



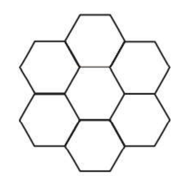
(a) 11 (b) 12
(c) 9 (d) 10

65. How many triangles are there in the following figure ?



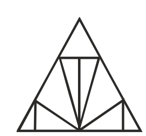
(a) 18 (b) 20
(c) 22 (d) 16

66. Six regular Hexagons of side 5 cm are joined together to form the figure given below. What is the perimeter of this figure?



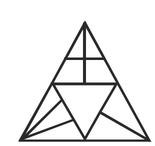
(a) 210 (b) 180
(c) 120 (d) 240

67. How many triangles are there in the given figure?



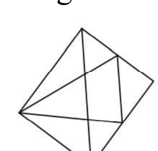
(a) 13 (b) 15
(c) 20 (d) 12

68. How many triangles are there in the given figure?



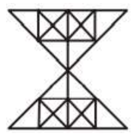
(a) 15 (b) 14
(c) 17 (d) 19

69. Find the number of triangles in the given figure :



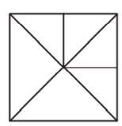
(a) 8 (b) 9
(c) 11 (d) 13

70. How many triangles are there in the given figure ?



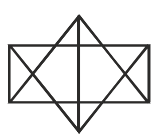
(a) 48 (b) 60
(c) 56 (d) 52

71. Find out the number of triangles in the figure given :



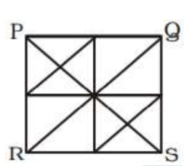
(a) 6 (b) 8
(c) 10 (d) 12

72. How many triangles are there in the given figure ?



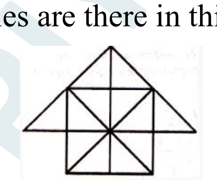
(a) 16 (b) 25
(c) 24 (d) 26

73. How many triangles are there in the following figure PQSR?



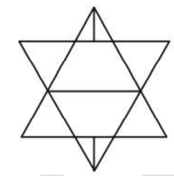
(a) 12 (b) 20
(c) 24 (d) 28

74. How many triangles are there in this figure ?



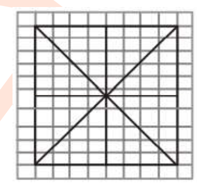
(a) 26 (b) 27
(c) 28 (d) 25

75. How many triangles are there in this figure ?



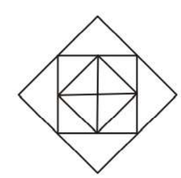
(a) 10 (b) 12
(c) 14 (d) 16

76. What is total number of triangles in the given figure ?



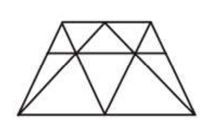
(a) 16 (b) 32
(c) 40 (d) 12

77. How many triangles are there in this figure ?



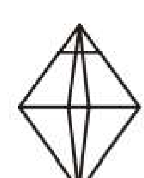
(a) 12 (b) 16
(c) 9 (d) 8

78. How many triangles are there in the given figure ?



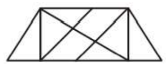
(a) 18 (b) 19
(c) 20 (d) 21

79. How many triangles are there in this geometric figure?



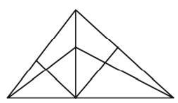
(a) 12 (b) 16
(c) 18 (d) 20

80. Find the number of triangles in the given figure.



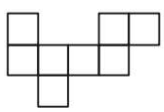
(a) 8 (b) 10
(c) 12 (d) 14

81. How many triangles are there in the following figure?



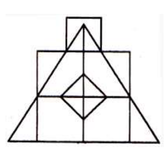
(a) 18 (b) 13
(c) 9 (d) 5

82. How many rectangles can you see in the figure?



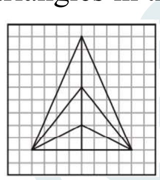
(a) 9 (b) 8
(c) 10 (d) 7

83. How many triangles are there in the given figure ?



(a) 18 (b) 19
(c) 20 (d) 21

84. Find the number of triangles in the given figure :

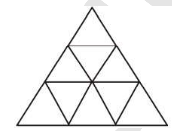


(a) 14 (b) 15
(c) 16 (d) 20

85. How many squares are there in this figure ?

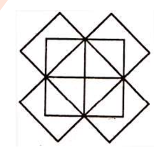
(a) 10 (b) 16
(c) 19 (d) 14

86. Find the number of triangles in the given figure :



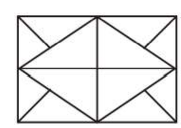
(a) 11 (b) 12
(c) 13 (d) 14

87. How many rectangles are there in the question figure?



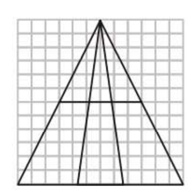
(a) 20 (b) 18
(c) 16 (d) 19

88. How many triangles are there in the following figure?



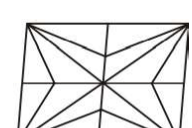
(a) 12 (b) 16
(c) 10 (d) 20

89. Find the number of triangles in the given figure.



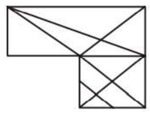
(a) 12 (b) 14
(c) 16 (d) 18

90. How many triangles are there in the given figure?



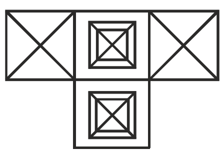
(a) 24 (b) 28
(c) 36 (d) 32

91. How many triangles can be found out from the following figure?



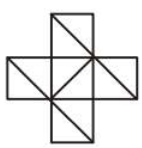
(a) 17 (b) 21
(c) 24 (d) 25

92. How many squares and triangles are there in the given figure?



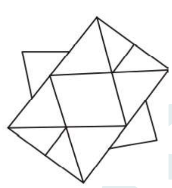
(a) 8,32 (b) 8,40
(c) 8,48 (d) 8,50

93. How many triangles are there in the given figure?



(a) 10 (b) 12
(c) 15 (d) 16

94. Find the number of triangles in the figure?

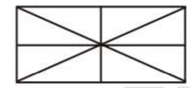


(a) 12 (b) 10
(c) 18 (d) 16

95. Find the number of triangles in the figure.

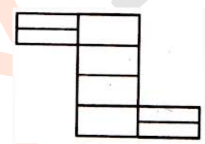
(a) 12 (b) 20
(c) 22 (d) 24

96. How many rectangles are there in the given figure ?



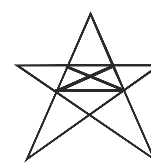
(a) 8 (b) 5
(c) 9 (d) 4

97. How many rectangles are there in the question figure?



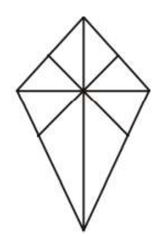
(a) 8 (b) 17
(c) 18 (d) 20

98. Find the number of triangles in the figure.



(a) 16 (b) 18
(c) 20 (d) 22 or more

99. How many triangles are there in the given figure?



(a) 14 (b) 15
(c) 16 (d) 18

100. Find the number of triangles in the figure.

(a) 12 (b) 18
(c) 22 (d) 26

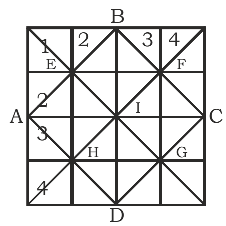
Answer Key

1	(c)	2	(c)	3	(b)	4	(a)	5	(c)	6	(b)	7	(d)	8	(d)	9	(c)	10	(d)
11.	(b)	12.	(a)	13	(c)	14	(d)	15	(c)	16	(d)	17	(c)	18	(d)	19	(d)	20	(b)
21.	(a)	22	(d)	23	(c)	24	(b)	25	(d)	26	(c)	27	(c)	28	(c)	29	(c)	30	(c)
31	(c)	32	(d)	33	(d)	34	(b)	35	(d)	36	(c)	37	(b)	38	(d)	39	(a)	40	(b)
41	(b)	42	(d)	43	(a)	44	(d)	45	(d)	46	(b)	47	(a)	48	(c)	49	(d)	50	(d)
51	(d)	52	(d)	53	(b)	54	(b)	55	(a)	56	(c)	57	(a)	58	(c)	59	(c)	60	(b)
61	(d)	62	(d)	63	(a)	64	(a)	65	(a)	66	(c)	67	(b)	68	(c)	69	(d)	70	(b)
71	(d)	72	(c)	73	(d)	74	(c)	75	(c)	76	(a)	77	(b)	78	(a)	79	(c)	80	(d)
81	(a)	82	(c)	83	(d)	84	(b)	85	(c)	86	(c)	87	(a)	88	(d)	89	(a)	90	(c)
91	(d)	92	(c)	93	(d)	94	(c)	95	(c)	96	(c)	97	(c)	98	(d)	99	(c)	100	(b)

★★★

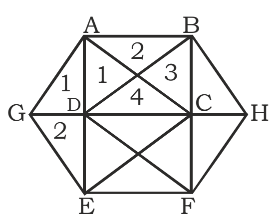
SOLUTION

1.



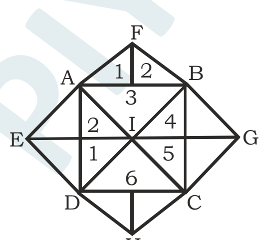
Basic squares : $(4\times4)+(3\times3)+(2\times2)+(1\times1) = 30$
Other squares : ABCD, AEIH, BFIE, CGIF, DHIG = 5
Total squares = 30+5 = 35

2.



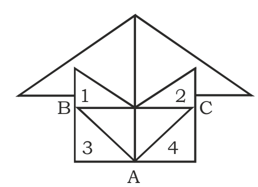
There are 4 original figures in the given figure:
Figure (1)- ABCD
Figure (2)- CDEF
Figure (3)- $\triangle$AEG
Figure (4)- $\triangle$BHF
Number of triangles in figure (1) & (2) = $2\times(4\times2) = 16$
Number of triangles in figure (3) & (4) = $2\times(1+2) = 6$
By connecting figure (1) & (2): $\triangle$ACE & $\triangle$BDF
By connecting figure (1), (2) & (3): $\triangle$GAC & $\triangle$CEG
By connecting figure (1), (2) & (4): $\triangle$BHD & $\triangle$HFD
$\therefore$ Total triangles = 16+6+2+2+2 = 28

3.



The figure given has 5 original figures:
Figure (1)- $\triangle$AFB
Figure (2)- $\triangle$BGC
Figure (3)- $\triangle$CDH
Figure (4)- $\triangle$ADE
Figure (5)- ABCD
Number of triangles in figure- (1),(2),(3) & (4) = $4\times(1+2) = 12$
Number of triangles in figure- (5) = $6\times2 = 12$
By connecting figure (4) & (5): $\triangle$AEI & $\triangle$EID
By connecting figure (2) & (5): $\triangle$BIG & $\triangle$CIG
$\therefore$ Total triangles = 12+12+2+2 = 28

4.



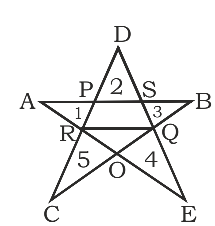
Number of triangles in $\triangle$ABC = (1+2) = 3
There are 4 triangles which have been numbered in the given figure.
Total triangles = 3+4 = 7

5. Total rectangles = 9 + 12 = 21
(See example-14 for detailed explanation)

6. Total triangles = 27
(See example-61 for detailed explanation)

7. Total squares = $(4\times4)+(3\times3)+(2\times2)+(1\times1) = 30$

8.



There are 5 small triangles which have been numbered in the given figure. Also this figure is made up of five lines (AB,BC,CD,DE,EA).

Each line will form one new triangle. Thus we will get 5 more triangles-

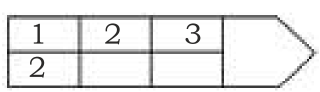
$\triangle$ABO, $\triangle$BCP, $\triangle$CDQ, $\triangle$DER & $\triangle$EAS

Also the line QR will give 4 new triangles : $\triangle$QRD, $\triangle$QRO, $\triangle$QRC & $\triangle$QRE

$\therefore$ Total triangles = 5+5+4 = 14

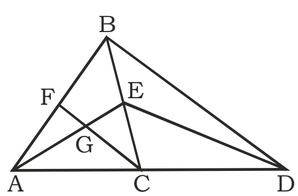
9. Total triangles = 28
(See example-43 for detailed explanation)

10.



Total rectangles = (1+2+3)×(1+2) = 18

11.



$\triangle$BCD will give (1+2) = 3 triangles

Number of triangles in $\triangle$ABC:

(i) Number of triangles in $\triangle$AFC: (1+2) = 3

(ii) 5 triangles: $\triangle$ECG, $\triangle$ACE, $\triangle$BFC, $\triangle$ABE, $\triangle$ABC

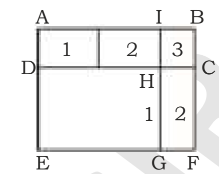
By connecting $\triangle$ABC & $\triangle$BCD : $\triangle$AED & $\triangle$ABD

$\therefore$ Total triangles = 3+8+2 = 13

12. Total triangles = 16
(See example-34 for detailed explanation)

13. Total triangles = 45
According to given options we have to mark option (C).
(See example-53 for detailed explanation)

14.



Number of rectangles in figure ABCD = 1×(1+2+3)=6

Number of rectangles in figure CDEF = 1× (1+2) = 3

3 rectangles : AEGI, BFGI, ABFE

Total rectangles = 6+3+3 = 12

According to given options we have to mark option (D).

15.

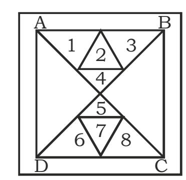
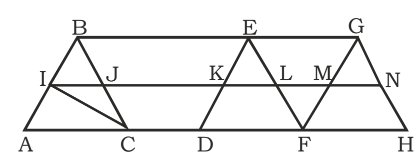


Figure ABCD will give 8 basic triangles = 2×4 = 8

Also there are 8 small triangles which have been numbered in the figure.

$\therefore$ Total triangles = 8+8 = 16

16.



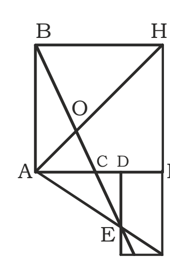
4 triangles : $\triangle$ABC, $\triangle$DEF, $\triangle$FGH & $\triangle$EFG

4 triangles : $\triangle$BJI, $\triangle$ELK, $\triangle$MNG & $\triangle$FML

Also the line IC will give 3 new triangles: $\triangle$ICJ, $\triangle$ICA, $\triangle$ICB

$\therefore$ Total triangles = 4+4+3 = 11

17.



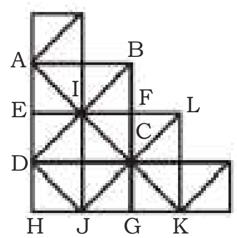
Number of triangles in $\triangle ABC$, $\triangle ADE$, $\triangle AGH$ & $\triangle EFG = 4\times(1+2) = 12$

Other triangles : $\triangle ABH$, $\triangle BHO$, $\triangle ABE$, $\triangle AOE$

$\therefore$ Total triangles = 12+4 = 16

According to given options we have to mark option (C).

18.

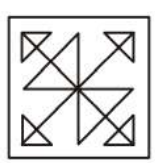


There are 10 small squares.

4 squares : ABCD, EFGH, IJKL & ICJD

$\therefore$ Total squares = 10+4 = 14

19.

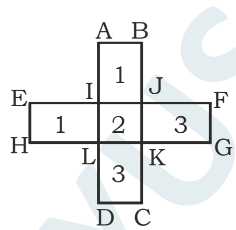


The basic figure "" will give 3 triangles.

There are six such figures.

$\therefore$ Total triangles = $6\times3 = 18$

20.



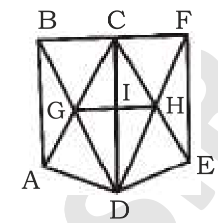
Number of rectangles in figure ABCD $= 1\times(1+2+3) = 6$

Number of rectangles in figure EFGH $= 1\times(1+2+3) = 6$

There is one common rectangle IJKL which has been counted twice, therefore we have to subtract it once.

$\therefore$ Total rectangles = 6+6–1 = 11

21.



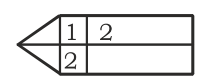
Number of basic triangles in figure ABCD & CDEF $= 2\times(2\times4) = 16$

4 triangles : $\triangle CIG$, $\triangle CIH$, $\triangle GID$ & $\triangle IDH$

3 triangles : $\triangle GCH$, $\triangle GDH$, $\triangle BDF$

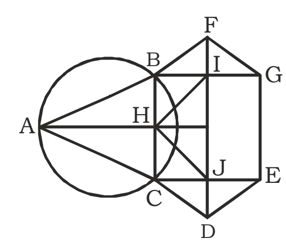
$\therefore$ Total triangles = 16+4+3 = 23

22.



Total rectangles $= (1+2)\times(1+2) = 9$

23.



Number of basic triangles in figure $\triangle ABC$, $\triangle BFG$, $\triangle CDE$ & $\triangle HIJ = 4\times(1+2) = 12$

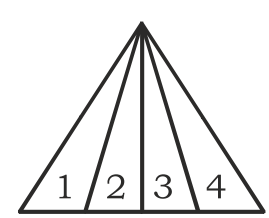
2 triangles : $\triangle BIH$ & $\triangle HJC$

$\therefore$ Total triangles = 12+2 = 14

24. Total triangles = 15

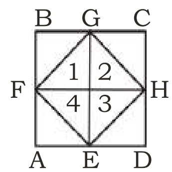
(See example-38 for detailed explanation)

25.



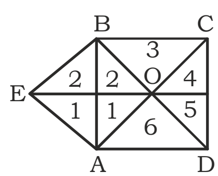
Total triangles = 1+2+3+4 = 10

26.



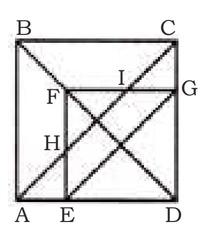
Number of triangles in figure EFGH = 2×4 = 8
4 triangles : △AEF, △BGF, △CHG & △EDH
∴ Total triangles = 8+4 = 12

27.



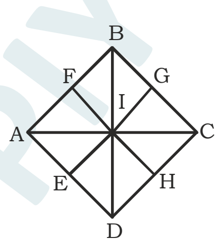
Number of triangles in ABCD = 2×6 = 12
Number of triangles in △EBA = 1+2 = 3
2 triangles will be formed by connecting both figures : △EBO & △EAO
∴ Total triangles = 12+3+2 = 17

28.



Number of triangles in ABCD = 2×4 = 8
Number of triangles in DEFG = 2×4 = 8
Number of triangles in △FHI = 1+2 = 3
2 triangles : △AEH & △ICG
∴ Total triangles = 8+8+3+2 = 21

29.



5 rhombuses : ABCD, AFIE, FBGI, IGCH, EIHD

30. Total triangles = 29
(See example-52 for detailed explanation)

31. There are 3 "⊞" such figures in the given figure. Each figure will form 5 squares.
∴ Total squares = 3×5 = 15
Here, no new squares are formed by connecting all the figures.

32.

1	2	3
2		
3		

Total squares = (3×3)+(2×2)+(1×1) = 14

33. Here we can easily count the number of circles in the given figure.
Total circles = 21

34.

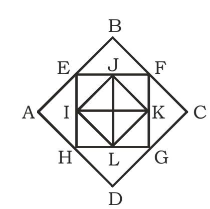
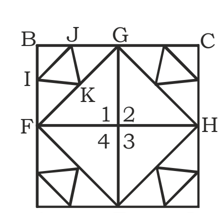


Figure EFGH will give 5 basic squares.
2 squares: ABCD & IJKL
Total squares = 5+2 = 7

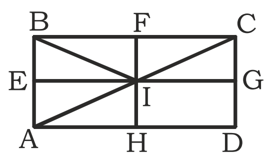
35. There are 2 "⊞" such figures in the given figure.
Each figure will form 5 squares.
∴ Total squares = 2×5 = 10

36.



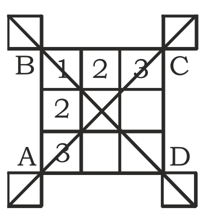
Number of triangles in figure EFGH = 4×2 = 8
Number of triangles in $\triangle$FGB = 5 [$\triangle$FGB, $\triangle$FIK, $\triangle$IBJ, $\triangle$JGK & $\triangle$IJK]
There are four figures like $\triangle$FGB.
$\therefore$ Number of triangles from these 4 figures = 4×5 = 20
Total triangles = 20+8 = 28

37.



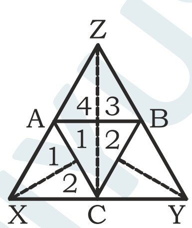
Number of triangles in $\triangle$ABI & $\triangle$BCI = 2×(1+2)=6
4 triangles: $\triangle$ABC, $\triangle$ADC, $\triangle$AHI & $\triangle$CGI
Total triangles = 6+4 = 10

38.



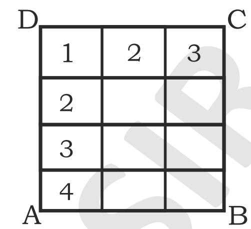
Number of squares in figure ABCD
= (3×3) + (2×2) + (1×1) = 14
Also, there are 4 other squares.
$\therefore$ Total squares = 14+4 = 18

39.

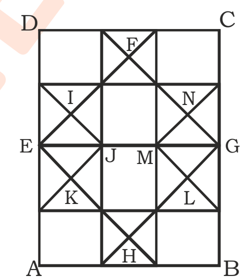


Number of triangles in ACBZ = 4×2 = 8
Number of triangles in $\triangle$XAC and $\triangle$YBC = 2 (1+2) = 6
3 triangles : $\triangle$XZC, $\triangle$YZC and $\triangle$XYZ
$\therefore$ Total triangles = 8+6+3 = 17

40.



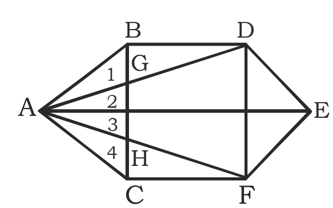
Basic squares in figure ABCD
= (4×3)+ (3×2)+ (2×1) = 20
Now,



3 squares : EFGH, EIJK & MNGL
$\therefore$ Total squares = 20+3 = 23

41. Here we can easily count the number of circles in the given figure.
Total circles = 6+6+1 = 13

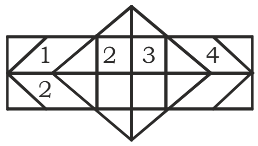
42.



Number of triangles in $\triangle$ABC = 1+2+3+4 = 10
Number of triangles in ADEF = 4×2 = 8
4 triangles : $\triangle$ABD, $\triangle$ACF, $\triangle$BDG & $\triangle$HFC
$\therefore$ Total triangles = 10+8+4 = 22

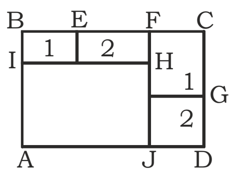
43. Total triangles = 44
(See example-47 for detailed explanation)

44.



Total rectangles = (1+2+3+4) × (1+2) = 30

45.



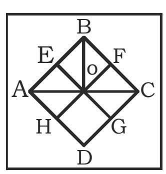
Number of rectangles in figure BFHI = 1×(1+2) = 3

Number of rectangles in figure FCDJ = 1×(1+2) = 3

3 rectangles : HIAJ, ABFJ, ABCD

∴ Total rectangles = 3+3+3 = 9

46.



Number of triangles in △ABO = (1+2) = 3

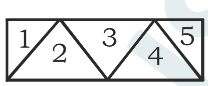
Number of triangles in △BCO = (1+2) = 3

2 triangles : △AHO, △CGO

2 triangles : △ABC, △ACD

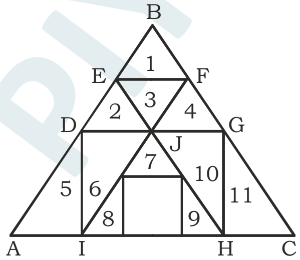
Total triangles = 3+3+2+2 = 10

47.



Total triangles = 5

48.

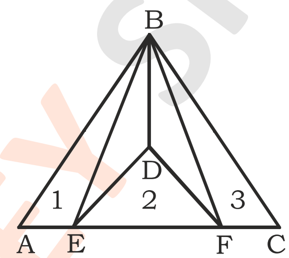


Here we can easily count 11 triangles in the given figure.

5 triangles : △BDG, △AHE, △ICF, △IHJ, △ABC

∴ Total triangles = 11+5 = 16

49.



Basic triangles in △ABC = (1+2+3) = 6

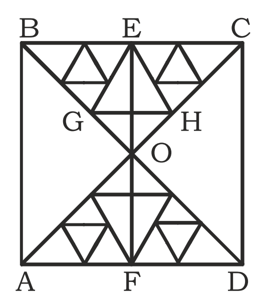
3 triangles : △DEF, △BDE & △BDF

∴ Total triangles = 6+3 = 9

50. Here we can easily see 4 blocks and 1 block is hidden.

Total blocks = 1+4 = 5

51.



Number of triangles in △BCO:

(i) Number of triangles in △BEG = 5

(ii) Number of triangles in △CEH = 5

(iii) Number of triangles in EGOH = 4×2 = 8

(iv) 3 triangles: △BEO, △CEO & △BCO

Total triangles = 5+5+8+3 = 21

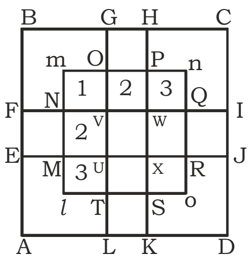
Similarly, there will be 21 triangles in △ADO

2 triangles : △ABO & △CDO

4 triangles : △ABC, △BCD, △CDA & △DAB

∴ Total triangles = 21+21+2+4 = 48

52.



Number of squares in lmno = (3×3)+(2×2)+(1×1) = 14

4 squares : EFNM, GHPO, IJRQ & KSTL

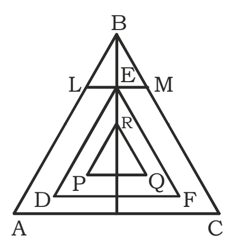
4 squares : FBGV, HCIW, JDKX & LAEU

4 squares : AFWK, BHXE, GCJU, IDLV

1 square : ABCD

∴ Total squares = 14+4+4+4+1 = 27

53.



Number of triangles in △PQR = (1+2) = 3

Similarly there will be 3-3 triangles in △ABC, △DEF & △BLM

Total triangles = 3×4 = 12

54. Here we can easily count the number of circles in the given figure.

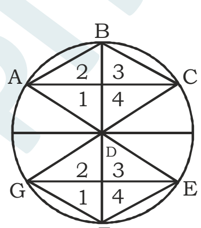
There are 3 big circles in the given figure and each big circle has 7 circles.

∴ Number of circles = 7×3 = 21

But there are two small circles which have been counted twice.

∴ Total circles = 21-2 = 19

55.

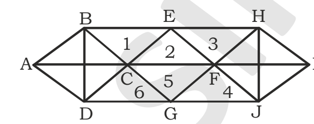


Number of triangles in figure ABCD = (4×2) = 8

Number of triangles in figure DEFG = (4×2) = 8

∴ Total triangles = 8+8 = 16

56.



Number of triangles in figure ABCD = 4×2 = 8

Number of triangles in figure FHIJ = 4×2 = 8

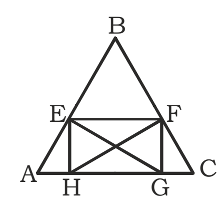
There are also 6 triangles which have been numbered in the figure.

2 triangles : △BGH & △DEJ

4 triangles : △BDG, △BDE, △HEJ & △GHJ

∴ Total triangles = 8+8+6+2+4 = 28

57.



Number of triangles in EFGH = 4×2 = 8

3 triangles : △AEH, △FCG, △EBF

3 triangles : △AEG, △CFH, △ABC

∴ Total triangles = 8+3+3 = 14

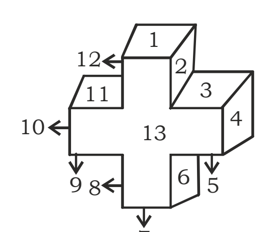
58. Here we can easily count the number of circles in the given figure.

Total circles = 17

59. Here we can easily count the number of circles in the given figure.

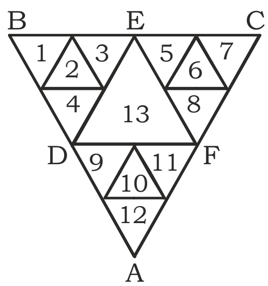
Total circles = 17

60.



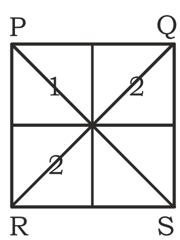
There are total 14 faces in this 3-D model.
Out of which 13 faces have been numbered in this model and 1 face will be back face which is hidden behind.

61.



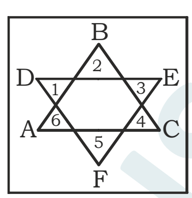
Here we can easily count 13 triangles which have been numbered in the given figure.
4 triangles : $\triangle$ADF, $\triangle$BDE, $\triangle$CEF & $\triangle$ABC
$\therefore$ Total triangles = 13+4 = 17

62.



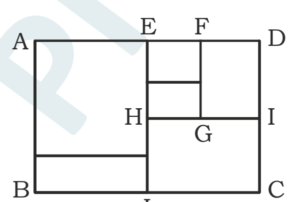
Total quadrilaterals = (1+2)×(1+2) = 9

63.



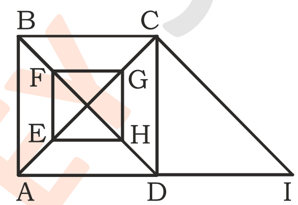
There are 6 small triangles which have been numbered.
2 triangles: $\triangle$ABC & $\triangle$DEF
$\therefore$ Total triangles = 6+2 = 8

64.



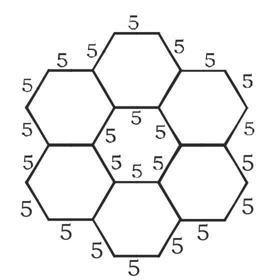
Number of rectangles in figure AEJB = 1× (1+2) = 3
Number of rectangles in figure EDCJ = 1× (1+2) = 3
Number of rectangles in figure EFGH = 1× (1+2) = 3
2 rectangles : FDIG & ABCD
$\therefore$ Total rectangles = 3+3+3+2 = 11

65.



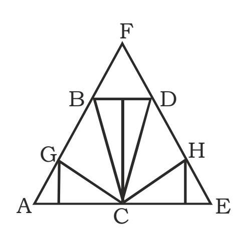
Number of triangles in figure ABCD = 4×2 = 8
Number of triangles in figure EFGH = 4×2 = 8
2 triangles : $\triangle$CDI & $\triangle$ACI
$\therefore$ Total triangles = 8+8+2 = 18

66.



Perimeter of this figure:
(i) Outer perimeter = 18×5 = 90 cm.
(ii) Inner perimeter = 6×5 = 30 cm.
Hence total perimeter = 90+30 = 120 cm.

67.



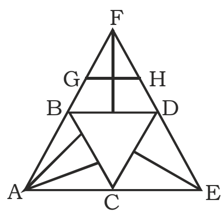
Number of triangles in $\triangle$ABC-
(i) Number of triangles in $\triangle$ACG = (1+2) = 3
(ii) 2 triangles: $\triangle$BCG & $\triangle$ABC
Similarly there will be 5 triangles in $\triangle$CDE.

Number of triangles in $\triangle$BCD = (1+2) = 3

2 triangles : $\triangle$BDF & $\triangle$AEF

Total triangles = 5+5+3+2 = 15

68.



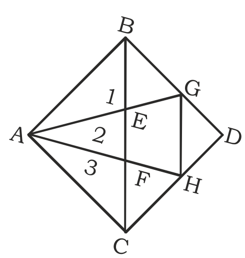
Number of triangles in $\triangle$ABC = (1+2+3) = 6

Number of triangles in $\triangle$FGH, $\triangle$FBD, $\triangle$CDE = 3× (1+2) = 9

2 triangles: $\triangle$BCD & $\triangle$AEF

Total triangles = 6+9+2 = 17

69.



Number of triangles in figure ABC = (1+2+3) = 6

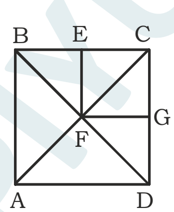
3 triangles : $\triangle$BEG, $\triangle$GDH & $\triangle$FHC

4 triangles : $\triangle$BCD, $\triangle$ABG, $\triangle$ACH & $\triangle$AGH

$\therefore$ Total triangles = 6+3+4 = 13

70. Total triangles = 60
(See example-54 for detailed explanation)

71.



Number of triangles in $\triangle$BCF = (1+2) = 3

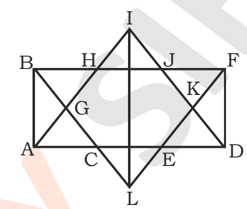
Number of triangles in $\triangle$CDF = (1+2) = 3

Number of triangles in $\triangle$ABD = (1+2) = 3

3 triangles : $\triangle$BCD, $\triangle$ACD, $\triangle$ABC

$\therefore$ Total triangles = 3+3+3+3 = 12

72.



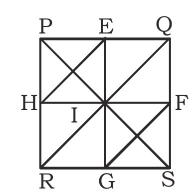
Number of triangles in $\triangle$ABC, $\triangle$DEF, $\triangle$HIJ, $\triangle$CEL, $\triangle$ADI & $\triangle$BFL = 6 (1+2) = 18

4 triangles : $\triangle$BGH, $\triangle$ABH, $\triangle$FJK & $\triangle$DFJ

2 triangles : $\triangle$GIL & $\triangle$IKL

Total triangles = 18+4+2 = 24

73.



Number of triangles in PEIH = 4×2 = 8

Number of triangles in IFSG = 4×2 = 8

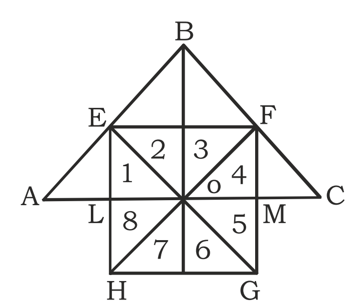
4 triangles : $\triangle$RHI, $\triangle$RGI, $\triangle$FQI & $\triangle$EQI

4 triangles : $\triangle$PQI, $\triangle$QSI, $\triangle$RSI & $\triangle$PRI

4 triangles : $\triangle$PQS, $\triangle$PRS, $\triangle$QRS & $\triangle$PRQ

$\therefore$ Total triangles : 8+8+4+4+4 = 28

74.



Number of triangles in figure EFGH = 8×2 = 16

Number of triangles in figure BEF = 1+2 = 3

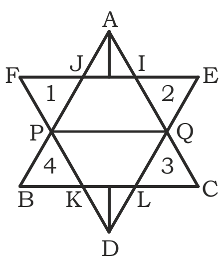
2 triangles : $\triangle$AEL & $\triangle$FCM

4 triangles : $\triangle$EBO, $\triangle$FBO, $\triangle$AEO & $\triangle$CFO

3 triangles : $\triangle ABO$, $\triangle BCO$ & $\triangle ABC$

$\therefore$ Total triangles = 16+3+2+4+3 = 28

75.



Number of triangles in $\triangle AIJ$ & $\triangle DKL$

= 2× (1+2) = 6

Also there are 4 triangles which have been numbered.

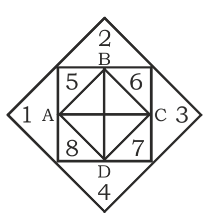
Line PQ will give 2 triangles: $\triangle APQ$ & $\triangle DPQ$

2 big triangles : $\triangle ABC$ & $\triangle DEF$

$\therefore$ Total triangles = 6+4+2+2 = 14

76. Total triangles = 8×2 = 16

77.

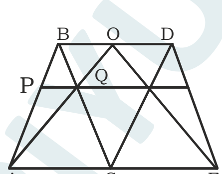


Number of triangles in figure ABCD = 2×4 = 8

Also there are 8 triangles which have been numbered in the figure.

$\therefore$ Total triangles = 8+8 = 16

78.



The given figure can be divided into three parts:

Figure (1) - $\triangle ABC$

Figure (2) - $\triangle CDE$

Figure (3) - $\triangle BCD$

Number of triangles in figure (1) :

(i) Number of triangles in $\triangle ABQ$ = (1+2) = 3

(ii) 2 triangles: $\triangle ACQ$ & $\triangle ABC$

Similarly there are 5 triangles in figure (2).

Also we can see that there are 5 triangles in figure (3) including $\triangle BCD$.

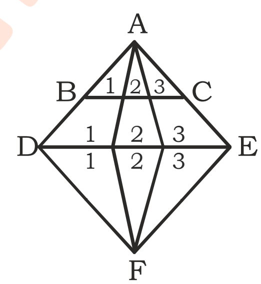
By connecting figure (1) & (3) – $\triangle ABO$

By connecting figure (2) & (3) – $\triangle DEO$

By connecting all the three figures – $\triangle AEO$

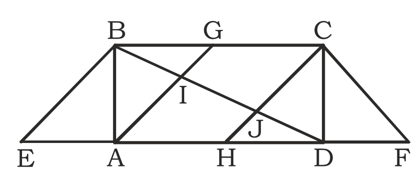
$\therefore$ Total triangles = 5+5+5+3 = 18

79.



Number of triangles in $\triangle ABC$, $\triangle ADE$ & $\triangle DEF$ = 3× (1+2+3) = 18

80.



Number of triangles in $\triangle ABG$ = (1+2) = 3

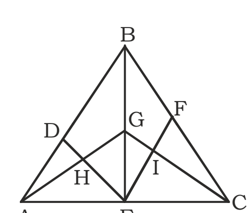
Number of triangles in $\triangle CDH$ = (1+2) = 3

4 triangles : $\triangle BCJ$, $\triangle AID$, $\triangle ABD$ & $\triangle BCD$

4 triangles : $\triangle ABE$, $\triangle CDF$, $\triangle EDB$ & $\triangle CFH$

$\therefore$ Total triangles = 3+3+4+4 = 14

81.



Number of triangles in $\triangle$ABE-

(i) Number of triangles in $\triangle$ADE = (1+2) = 3

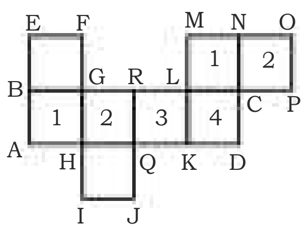
(ii) 5 triangles : $\triangle$GHE, $\triangle$AGE, $\triangle$DBE, $\triangle$ABG & $\triangle$ABE

Similarly there are 8 triangles in $\triangle$BCE.

By connecting both figures : $\triangle$AGC & $\triangle$ABC

$\therefore$ Total triangles = 8+8+2 = 18

82.



Number of rectangles in figure ABCD =1×(1+2+3+4)=10

Number of rectangles in figure LMOP = 1× (1+2) = 3

2 rectangles : BEFG, HIJQ

3 rectangles : AEFH, GIJR, DKMN

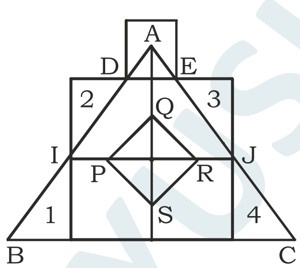
Total rectangles = 10+3+2+3 = 18

In the given figure there are 8 squares which have been included in counting of total rectangles.

Hence only rectangles = 18-8 = 10

According to given options we have to mark option (C).

83.



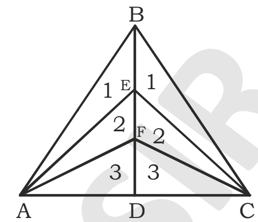
Number of triangles in $\triangle$ADE, $\triangle$AIJ & $\triangle$ABC = 3 × (1+2) = 9

Number of triangles in figure PQRS = 4×2 = 8

Also there are 4 triangles which have been numbered in the given figure.

$\therefore$ Total triangles = 9+8+4 = 21

84.

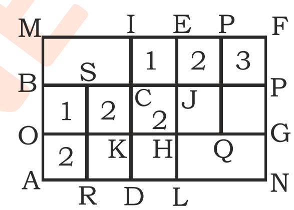


Number of triangles in $\triangle$ABD & $\triangle$BCD = 2×(1+2+3) = 12

3 triangles : $\triangle$ABC, $\triangle$AEC & $\triangle$AFC

$\therefore$ Total triangles = 12+3 = 15

85.



Number of squares in figure ABCD = (2×2)+(1×1) = 5

Number of squares in figure IFGK = (3×2)+(2×1) = 8

4 squares: KHLD, IKOM, JLRS, JLNP

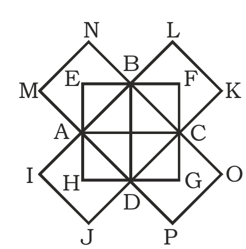
2 squares: AMEL, FNDI

Total squares = 5+8+4+2 = 19

86. Total triangles = 13

(See example-62 for detailed explanation)

87.



This question can be solved easily if we divide the given question figure in the following way:

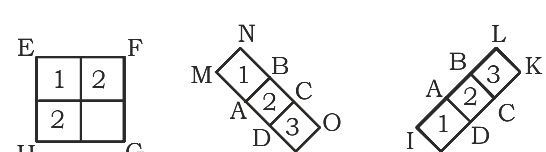
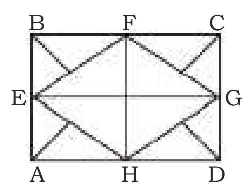


Figure (1) Figure (2) Figure (3)

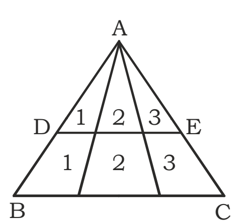
Number of rectangles in figure (1) = (1+2) × (1+2) = 9
Number of rectangles in figure (2) = 1× (1+2+3) = 6
Number of rectangles in figure (3) = 1× (1+2+3) = 6
There is one common rectangle ABCD in figure (2) & (3) which has been counted twice and therefore we have to subtract it.
∴ Total rectangles = 9+6+6-1 = 20

88.



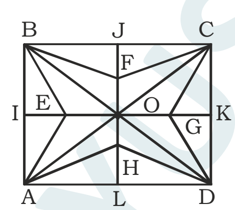
Number of triangles in $\triangle$BEF, $\triangle$FCG, $\triangle$GDH and $\triangle$AEH = 4×(1+2) = 12
Number of triangles in EFGH = 2×4 = 8
∴ Total triangles = 12+8 = 20

89.



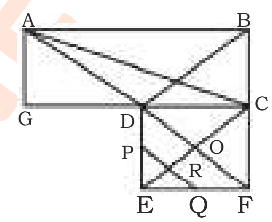
Number of triangles in $\triangle$ABC & ADE
= 2 (1+2+3) = 12

90.



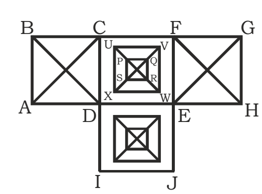
The given figure can be divided into following parts:
Figure (1)- $\triangle$ABO
Figure (2)- $\triangle$BCO
Figure (3)- $\triangle$CDO
Figure (4)- $\triangle$ADO
Number of triangles in $\triangle$ABO-
(i) In $\triangle$ABE : (1+2) = 3
(ii) In $\triangle$ABO : (1+2) = 3
(iii) 2 triangles : $\triangle$BEO & $\triangle$AEO
Total = 3+3+2 = 8
Similarly there will be 8-8 triangles in other figures.
∴ Number of triangles in four figures = 4×8 = 32
4 triangles : $\triangle$ABC, $\triangle$BCD, $\triangle$CDA & $\triangle$DAB
Total triangles = 32+4 = 36

91.



Number of triangles in figure ABCD = 4×2 = 8
Number of triangles in figure CDEF:
(i) Basic triangles = 4×2 =8
(ii) Line PQ will give 3 triangles : $\triangle$EPR, $\triangle$EQR & $\triangle$EPQ
2 triangles : $\triangle$ADG, $\triangle$ACG
4 triangles : $\triangle$ACO, $\triangle$ACF, $\triangle$BDF & $\triangle$ABF
∴ Total triangles = 8+8+3+2+4 = 25

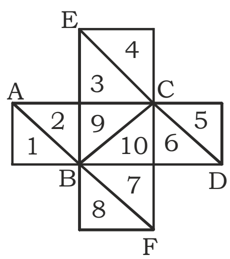
92.



From figure ABCD-
(i) Number of triangles : (4×2) = 8
(ii) Number of square : 1
Similarly we will get 8 triangles & 1 square from figure EFGH.
From figure CDEF-
(i) Number of triangles :
(a) In figure PQRS = 4×2 = 8
(b) In figure UVWX = 4×2 = 8
Total = 8+8 = 16

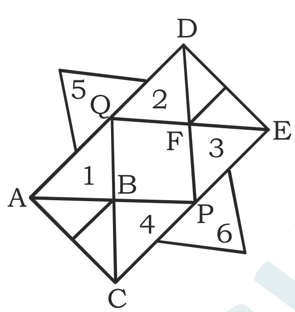
(ii) Number of squares: 3 [CDEF, UVWX, PQRS]
Similarly we will get 16 triangles & 3 squares from figure DEJI.
Now, Total triangles = (8×2) + (16×2) = 48
Total squares = (1×2) + (3×2) = 8

93.



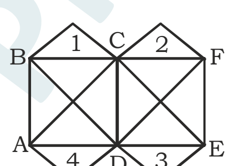
Here we can easily count 10 triangles which have been numbered in the given figure.
4 triangles : △ABC, △BCD, △EBC & △BCF
2 triangles : △EBD & △FAC
∴ Total triangles = 10+4+2 = 16

94.



Number of triangles in △ABC & △DEF = 2× (1+2)=6
Also there are 6 triangles which have been numbered in the given figure.
6 triangles : △ACP, △ACQ, △DEP, △DEQ, △ADP, △CEQ
∴ Total triangles = 6+6+6 = 18

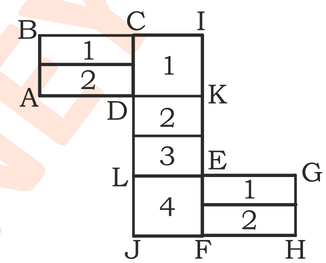
95.



Number of triangles in ABCD & CDEF = 2×(4×2) = 16
Also there are 4 triangles which have been numbered in the given figure.
2 triangles : △BDF & △ACE
∴ Total triangles = 16+4+2 = 22

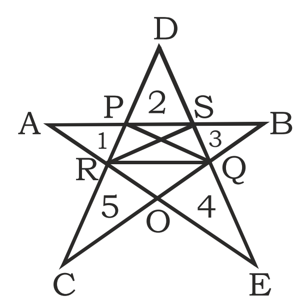
96. Total rectangles = (1+2) × (1+2) = 9

97.



Number of rectangles in ABCD = 1× (1+2) = 3
Number of rectangles in EFHG = 1× (1+2) = 3
Number of rectangles in CIFJ = 1× (1+2+3+4) = 10
2 rectangles : ABIK & LGHJ
∴ Total rectangles = 3+3+10+2 = 18

98.



There are 5 small triangles which have been numbered in the given figure. Also this figure is made up of five lines (AB,BC,CD,DE,EA).
Each line will form one new triangle. Thus we will get 5 more triangles-
△ABO, △BCP, △CDQ, △DER & △EAS
Number of triangles in figure PSQR = 4×2 = 8
Also the line QR will give 4 new triangles : △QRD, △QRO, △QRC & △QRE

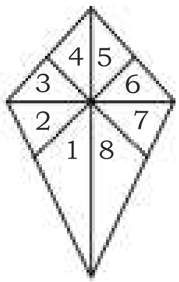
Line PQ will give 3 new triangles: $\triangle CPQ$, $\triangle DPQ$, $\triangle BPQ$

Line RS will give 3 new triangles: $\triangle ARS$, $\triangle DRS$, $\triangle ERS$

$\therefore$ Total triangles = 5+5+8+4+3+3 = 28

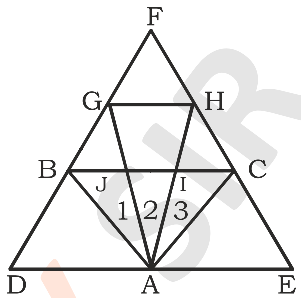
According to given options we have to mark option (D).

99.



Total triangles = 8×2 = 16

100.



Number of triangles in $\triangle ABC = (1+2+3) = 6$

6 triangles : $\triangle ABD$, $\triangle ACE$, $\triangle FGH$, $\triangle FBC$, $\triangle BGJ$ & $\triangle IHC$

4 triangles : $\triangle ABG$, $\triangle ACH$, $\triangle ADG$ & $\triangle AHE$

2 triangles : $\triangle AGH$ & $\triangle DEF$

$\therefore$ Total triangles = 6+6+4+2 = 18